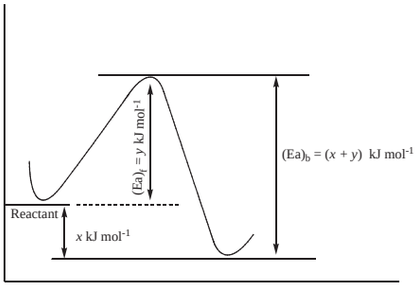
10. option(d)



$(x+y)\,\text{kJmol}^{-1}$

$(x+y)\times 10^{3}\ \text{Jmol}^{-1}$

11. option(c)

$T_1 = 200K\ ;\ k = k_1$

$T_2 = 400K\ ;\ k = k_2 = 2k_1$

$$\log\left(\frac{k_2}{k_1}\right) = \frac{2.303\ E_a}{R}\left(\frac{T_2 - T_1}{T_1\ T_2}\right)$$

$$\log\left(\frac{2\ k_1}{k_1}\right) = \frac{2.303\ E_a}{8.314\ J\ K^{-1}mol^{-1}}\left(\frac{400\ K - 200K}{200K\ \times\ 400K}\right)$$

$$E_a = \frac{0.3010\ \times\ 8.314\ J\ mol^{-1}\times\ 200\ \times\ 400}{2.303\ \times\ 200}$$

$E_a = 434.65\ J\ mol^{-1}$

12. option(b)

$$k = \left(\frac{2.303}{t}\right)\log\left(\frac{[A_0]}{[A]}\right)$$

$$2.303\ \times\ 10^{-2}\ hour^{-1} = \left(\frac{2.303}{1806\ min}\right)\log\left(\frac{0.25}{[A]}\right)$$

$$\left(\frac{2.303\ \times\ 10^{-2}\ hour^{-1}\times 1806\ min}{2.303}\right) = \log\left(\frac{0.25}{[A]}\right)$$

Answers.indd 246 3/29/2019 6:44:01 PM